

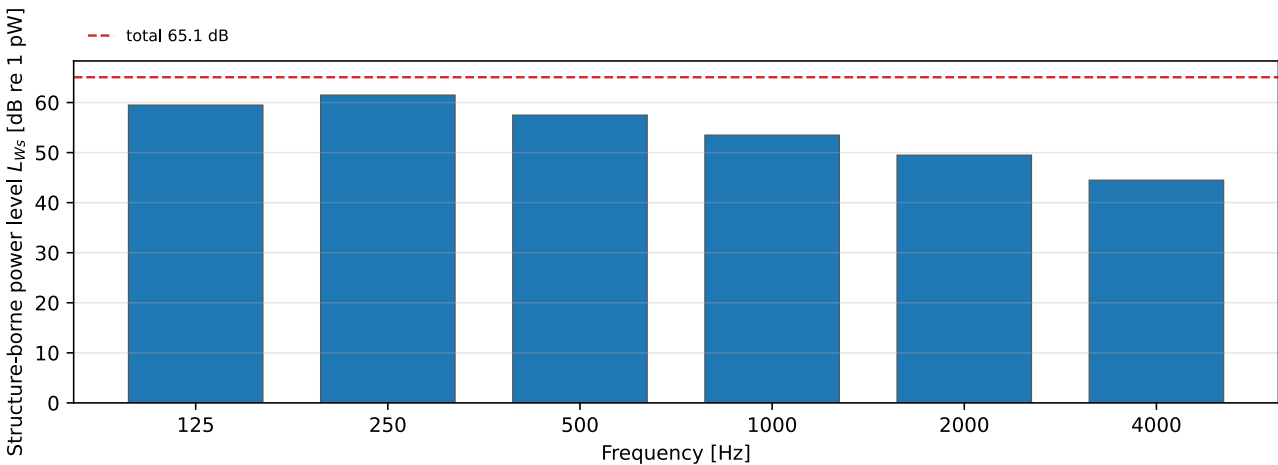
# Structure-borne sound power determination

Determination of the structure-borne sound power injected into the reception plate by building service equipment (EN 15657:2018 reception-plate method, Formula 14).

|                         |                                                 |                        |                                                            |
|-------------------------|-------------------------------------------------|------------------------|------------------------------------------------------------|
| Client:                 | Example building services contractor            | Noise source:          | Circulation pump (wall-mounted)                            |
| Test environment:       | Reception-plate test rig (heavy concrete plate) | Instrumentation:       | Piezoelectric accelerometer array (ISO 16063-21), s/n 0042 |
| Temperature [°C]:       | 21                                              | Relative humidity [%]: | 45                                                         |
| Ambient pressure [kPa]: | 101.1                                           | Date of test:          | 2026-07-21                                                 |

Octave-band structure-borne sound power levels

| f [Hz] | L <sub>v</sub> [dB] | L <sub>WS</sub> [dB] |
|--------|---------------------|----------------------|
| 125    | 70.0                | 59.5                 |
| 250    | 72.0                | 61.5                 |
| 500    | 68.0                | 57.5                 |
| 1000   | 64.0                | 53.5                 |
| 2000   | 60.0                | 49.5                 |
| 4000   | 55.0                | 44.5                 |



Structure-borne sound power level  $L_{WS} = 65.1$  dB re 1 pW

Plate mass per area  $m = 230.0$  kg/m<sup>2</sup>  
Reception-plate area  $S = 7.00$  m<sup>2</sup>  
Plate-specific level (Formula 14); convert to the source quantity before EN 12354-5.

$L_{WS} = 10 \lg(2\pi f \eta m S / (f_0 m_0 S_0)) + L_v - 60$  dB re 1 pW (EN 15657:2018 Formula 14), with the references  $f_0 = 1$  Hz,  $m_0 = 1$  kg,  $S_0 = 1$  m<sup>2</sup>, the plate loss factor  $\eta = 2,2/(f T_s)$  (Formula 13) and the spatial mean velocity level  $L_v = 10 \lg((1/N) \sum 10^{L_{v_i}/10})$  re  $v_0 = 10^{-9}$  m/s (Formula 12).

Formula 14 gives the power injected into this particular reception plate, not a plate-independent source descriptor: convert it to the equivalent blocked force level (Formula 15) and the characteristic reception-plate power level  $L_{WS,n}$  (Formula 17), then apply the EN 12354-5:2009 Annex I mobility correction, before predicting an installed level.

**Laboratory:** Phonometry reference example

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**Date:** 2026-07-21

**Notes:** Structure-borne sound power injected into the reception plate (EN 15657:2018 reception-plate method, Formula 14).

Operator: phonometry    Signature: \_\_\_\_\_

■ The results relate only to the tested specimen.

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